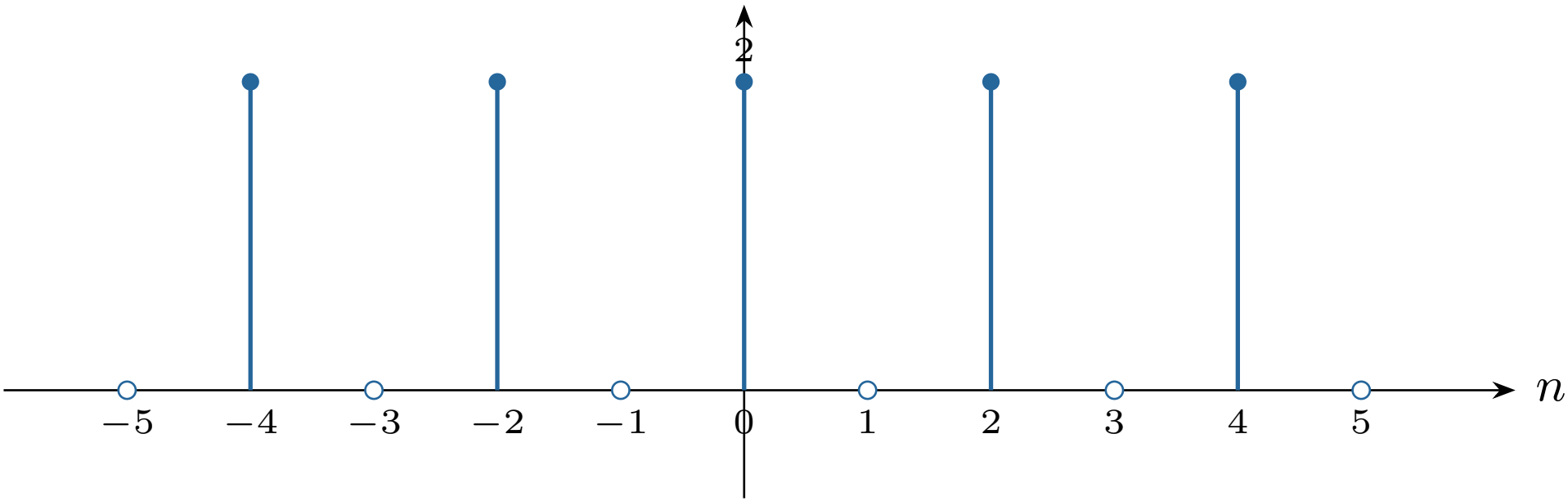
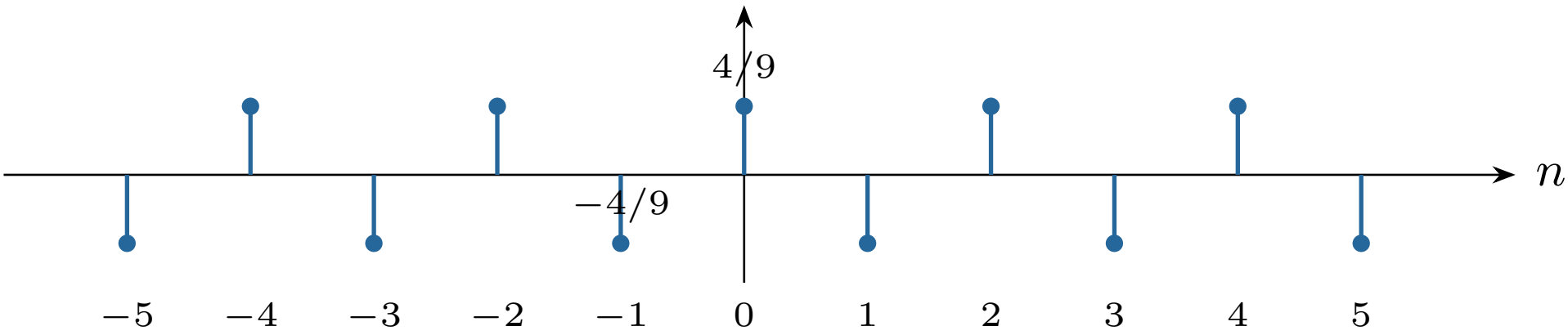
$x[n] = 1 + (-1)^n$



$y[n] = \frac{4}{9}(-1)^n$